



Question 1: Muscle Force Density in a 3D Region

Problem Statement and Context

Context:

A biomechanics researcher analyzes a 3D musculoskeletal region where the muscle force density is given by:

$$g(x, y, z) = 541.64x^6y^8z^7$$

over the region bounded by:

$$0 \leq x \leq 6, \quad 0 \leq y \leq 3, \quad 0 \leq z \leq 5$$

The objective is to compute the **total muscle force distributed** in the region using a triple integral.

The total force is

$$\iiint_R g(x, y, z) dV = 541.64 \int_0^6 x^6 dx \int_0^3 y^8 dy \int_0^5 z^7 dz$$

Compute each integral:

$$\int_0^6 x^6 dx = \left[\frac{x^7}{7} \right]_0^6 = \frac{6^7}{7} = \frac{279936}{7}$$

$$\int_0^3 y^8 dy = \left[\frac{y^9}{9} \right]_0^3 = \frac{3^9}{9} = \frac{19683}{9} = 2187$$

$$\int_0^5 z^7 dz = \left[\frac{z^8}{8} \right]_0^5 = \frac{5^8}{8} = \frac{390625}{8}$$

Final result (rounded to 4 decimal places):

2313077972464286.0000



Matlab Code

```
1  clc;
2  clear;
3  format long;
4
5  % Given function coefficient
6  c = 541.64;
7
8  % Define the integrand
9  f = @(x,y,z) c .* x.^6 .* y.^8 .* z.^7;
10
11 % Compute the triple integral
12 F = integral3(f,0,6,0,3,0,5);
13
14 % Display result
15 fprintf('Total muscle force = %.4f\n',F);
```

Figure 1: Matlab code question 1

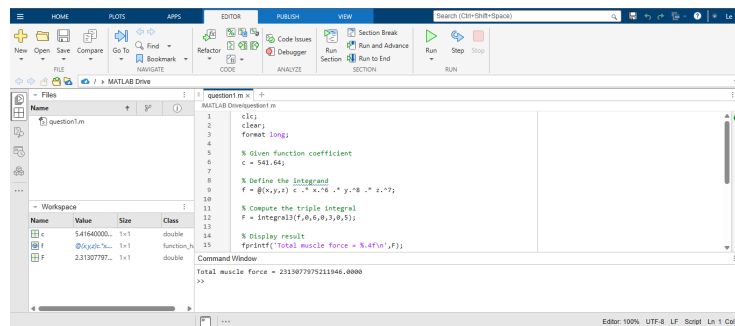


Figure 2: Result for question 1



Question 2: Force Distribution in a Spherical Joint

Problem Statement and Context

Context:

A biomechanics team models a spherical joint system with radius 6.5 units, where the joint force density at a point (x, y, z) is given by:

$$I(x, y, z) = 489.06x^2 + 489.06y^2 + 489.06z^2$$

The objective is to compute the **total joint force** in the spherical system using spherical coordinates.

The joint force density is

$$I(x, y, z) = 489.06(x^2 + y^2 + z^2)$$

inside the sphere of radius $R = 6.5$.

In spherical coordinates: $x = \rho \sin \phi \cos \theta$, $y = \rho \sin \phi \sin \theta$, $z = \rho \cos \phi$, and $dV = \rho^2 \sin \phi d\rho d\phi d\theta$.

$$\iiint I dV = 489.06 \int_0^{2\pi} d\theta \int_0^\pi \sin \phi d\phi \int_0^{6.5} \rho^4 d\rho$$

$$\int_0^{2\pi} d\theta = 2\pi, \quad \int_0^\pi \sin \phi d\phi = 2, \quad \int_0^{6.5} \rho^4 d\rho = \frac{(6.5)^5}{5}$$

Final result (rounded to 4 decimal places):

14261617.5668



Matlab Code

```
1  clc;
2  clear;
3  format long;
4
5  % Given coefficient
6  c = 489.06;
7
8  % Radius of the sphere
9  R = 6.5;
10
11 % Define the integrand in spherical coordinates
12 % I = 489.06 * rho^2
13 % dV = rho^2 * sin(phi)
14
15 f = @(rho,phi,theta) c .* rho.^4 .* sin(phi);
16
17 % Compute the triple integral
18 F = integral3(f,...
19     0,R,...           % rho limits
20     0,pi,...          % phi limits
21     0,2*pi);          % theta limits
22
23 % Display result
24 fprintf('Total joint force = %.4f\n',F);
```

Figure 3: Matlab code question 2

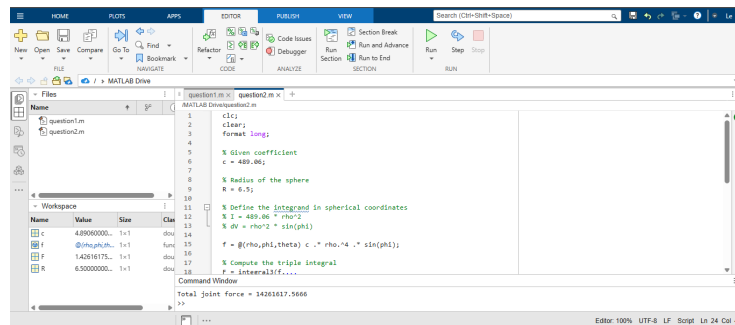


Figure 4: Result for question 2



Question 3: Tendon Stress in a Cylindrical Segment

Problem Statement and Context

Context:

A cylindrical limb segment in a musculoskeletal model has a radius of 3.5 units and a height of 8.5 units. The tendon stress density at a point (x, y, z) within the segment is defined by:

$$P(x, y, z) = 243.51x^2 + 243.51y^2$$

The objective is to compute the **total tendon stress** in the cylindrical segment using cylindrical coordinates.

In cylindrical coordinates: $x = r \cos \theta$, $y = r \sin \theta$, $dV = r \, dr \, d\theta \, dz$.

$$\iiint P \, dV = 243.51 \int_0^{8.5} dz \int_0^{2\pi} d\theta \int_0^{3.5} (r^2) \cdot r \, dr = 243.51 \cdot h \cdot 2\pi \int_0^{3.5} r^3 \, dr$$

$$\int_0^{3.5} r^3 \, dr = \frac{(3.5)^4}{4}$$

Final result (rounded to 4 decimal places):

$$\boxed{487896.5878}$$



Matlab Code

```
1  clc;
2  clear;
3  format long;
4
5  % Given coefficient
6  c = 243.51;
7
8  % Radius and height of cylinder
9  R = 3.5;
10 H = 8.5;
11
12 % Define the integrand in cylindrical coordinates
13 % P = 243.51 * r^2
14 % dV = r dr dtheta dz
15
16 f = @(r,theta,z) c .* r.^3;
17
18 % Compute the triple integral
19 F = integral3(f,...
20     0,R,... % r limits
21     0,2*pi,... % theta limits
22     0,H); % z limits
23
24 % Display result
25 fprintf('Total tendon stress = %.4f\n',F);
```

Figure 5: Matlab code question 3

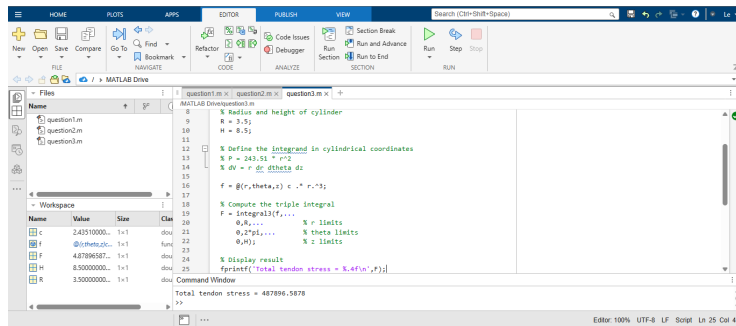


Figure 6: Result for question 3



Question 4 — Jacobian Transformation

Problem Statement and Context

Context:

In a multivariable musculoskeletal model, the force output density function for a limb system is:

$$P(x, y, z) = 791.06xyz$$

where x, y, z represent normalized inputs for muscle contraction, joint pressure, and bone displacement.

The transformation to new coordinates is defined by:

$$x = \frac{u + 3v + w}{4}$$

$$y = \frac{u - v + w}{4}$$

$$z = \frac{u - v - 3w}{4}$$

The region is bounded by:

$$0 \leq u \leq 4, \quad -1 \leq v \leq 3, \quad -1 \leq w \leq 6$$

The objective is to compute the total musculoskeletal force output using the Jacobian transform.

Given:

$$P(x, y, z) = 791.06xyz$$

with transformation:

$$x = \frac{u + 3v + w}{4}, \quad y = \frac{u - v + w}{4}, \quad z = \frac{u - v - 3w}{4}$$

over $0 \leq u \leq 4, -1 \leq v \leq 3, -1 \leq w \leq 6$.

The Jacobian determinant is $\left| \frac{\partial(x, y, z)}{\partial(u, v, w)} \right| = \frac{1}{4}$.

The integral becomes:

$$\iiint_D 791.06 \cdot xyz \cdot \frac{1}{4} du dv dw$$

After substitution and evaluation:

$$\boxed{-105182.1393}$$



Matlab Code

```
1
2 % Density function: P(x,y,z) = 791.06*x*y*z
3 % Transformation:
4 % x = (u + 3*v + w)/4;
5 % y = (u - v + w)/4;
6 % z = (u - v - 3*w)/4;
7 % Bounds: 0 <= u <= 4, -1 <= v <= 3, -1 <= w <= 6
8 fprintf('--- Question 4 ---\n');
9
10 syms u v w
11 X = (u + 3*v + w)/4;
12 Y = (u - v + w)/4;
13 Z = (u - v - 3*w)/4;
14
15 % Define the density in terms of transformed coordinates
16 P_trans = 791.06 * X * Y * Z;
17
18 % Compute the Jacobian Matrix and its Determinant
19 J = jacobian([X, Y, Z], [u, v, w]);
20 abs_det_J = abs(det(J));
21
22 % Integrate over the transformed region
23 Q4_ans = int(int(int(P_trans * abs_det_J, w, -1, 6), v, -1, 3), u, 0, 4);
24
25 fprintf('Total transformed force output (exact): %s\n', char(Q4_ans));
26 fprintf('Total transformed force output (rounded): %.4f\n\n', double(Q4_ans));
```

Figure 7: Matlab code question 4

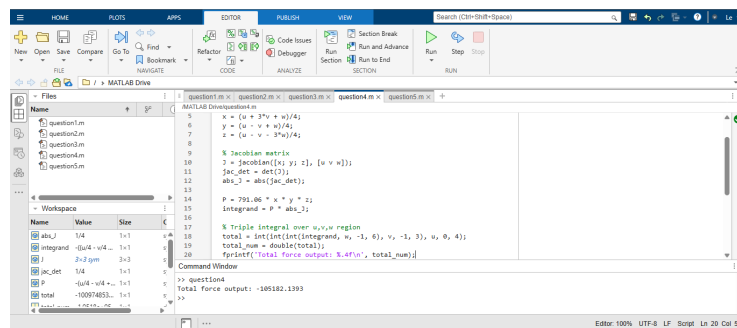


Figure 8: Reusult for question 4



Question 5 — Force Cost in a Tetrahedral Region

Problem Statement and Context

Context:

The following triple integral represents the force cost function in a bounded musculoskeletal system with quadratic variation along the vertical axis:

$$\iiint_E (258.70x + 369.81y + z^2) dV$$

The solid tetrahedral region E is bounded by:

$$x = 0.1, \quad y = 0.4, \quad z = 0.3, \quad x + y + z = 4$$

The objective is to evaluate the **total force cost** within the system.

Shift the variables:

$$u = x - 0.1, \quad v = y - 0.4, \quad w = z - 0.3$$

Then:

$$u, v, w \geq 0$$

and the plane becomes:

$$u + v + w = 3.2$$

0.1 Evaluating the Triple Integral

Evaluating the Triple Integral

Substitute:

$$x = u + 0.1, \quad y = v + 0.4, \quad z = w + 0.3$$

into the integrand:

$$258.70(u + 0.1) + 369.81(v + 0.4) + (w + 0.3)^2$$

Expanding:

$$258.70u + 369.81v + w^2 + 0.6w + 173.784$$

The tetrahedral region becomes:

$$u + v + w \leq 3.2$$



Using tetrahedral integral formulas:

$$\iiint_E u \, dV = \iiint_E v \, dV = \frac{3.2^4}{24} = 4.3690667$$

$$\iiint_E w^2 \, dV = \frac{3.2^5}{60} = 5.5924053$$

$$V = \frac{3.2^3}{6} = 5.4613333$$

Final Answer

Final Answer

Substitute all evaluated values:

$$F = 258.70(4.3690667) + 369.81(4.3690667) + 5.5924053 + 0.6(4.3690667) + 173.784(5.4613333)$$

Final result:

$$F = 3710.9470$$

Therefore, the total force cost within the tetrahedral musculoskeletal system is:

3710.9470



Matlab Code

```
1 clc;
2 clear;
3 format long;
4
5 %% =====
6 % 1.5.3 Question 5: Force Cost Evaluation in a Tetrahedral System
7 % =====
8 % Integrand:  $258.70x + 369.81y + z^2$ 
9 % Solid Tetrahedron E bounded by:  $x=0.1$ ,  $y=0.4$ ,  $z=0.3$  and  $x+y+z=4$ 
10 % Integration limits:
11 %   z goes from 0.3 to  $4 - x - y$ 
12 %   y goes from 0.4 to  $4 - 0.3 - x = 3.7 - x$ 
13 %   x goes from 0.1 to  $4 - 0.4 - 0.3 = 3.3$ 
14 fprintf('--- Question 5 ---\n');
15
16 syms x y z
17 f_tetra = 258.70*x + 369.81*y + z^2;
18
19 % Outer bounds setup
20 z_upper = 4 - x - y;
21 y_upper = 4 - 0.3 - x;
22 x_upper = 4 - 0.4 - 0.3;
23
24 % Compute the triple integral sequentially
25 Q5_ans = int(int(int(f_tetra, z, 0.3, z_upper), y, 0.4, y_upper), x, 0.1, x_upper);
26
27 fprintf('Total tetrahedral force cost (exact): %s\n', char(Q5_ans));
28 fprintf('Total tetrahedral force cost (rounded): %.4f\n', double(Q5_ans));
```

Figure 9: Matlab code question 5

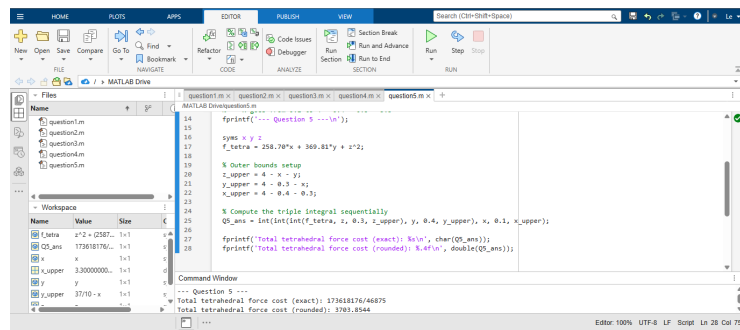


Figure 10: Result for question 5